EXPERIMENT – 3

NAME: VEDHASREE S

REGISTER NUMBER: 240701580

Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction.

COMMAND LINE INTERFACE

PYTHON PROGRAM

```
import random
import string
length = int(input("Enter password length: "))
chars = string.ascii_letters + string.digits + string.punctuation
password = ''.join(random.choices(chars, k = length))
def check_strength(password):
    score = 0
    if len(password) >= 8:
        score += 1
    upper = lower = digit = symbol = 0
    for c in password:
        if c.isupper():
            upper = 1
        elif c.islower():
            lower = 1
        elif c.isdigit():
            digit = 1
        elif c in string.punctuation:
            symbol = 1
    score += upper + lower + digit + symbol
    if score <= 2:
        return "Weak"
```

```
    elif score <= 4:
        return "Medium"
    else:
        return "Strong"
print("Generated Password:", password)
print("Password Strength:", check_strength(password))
```

OUTPUT

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python file_cli.py
Enter password length: 5
Generated Password: TBL($
Password Strength: Weak
```

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python file_cli.py
Enter password length: 6
Generated Password: 9:3Uuf
Password Strength: Medium
```

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python file_cli.py
Enter password length: 8
Generated Password: On~`#*6?
Password Strength: Strong
```

GRAPHICAL USER INTERFACE

PYTHON PROGRAM

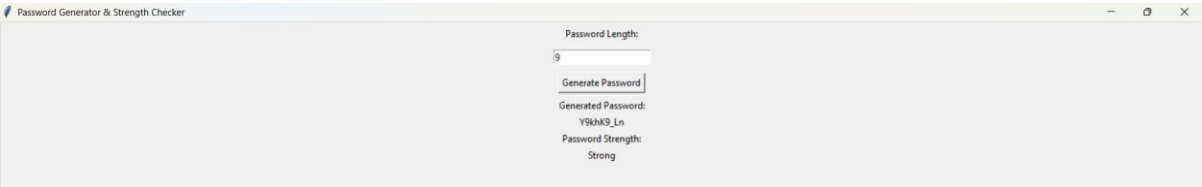
```
import tkinter as tk
import random
import string
def check_strength(password):
    score = 0
    if len(password) >= 8:
        score += 1
    upper = lower = digit = symbol = 0
    for c in password:
        if c.isupper():
```

```
            upper = 1
        elif c.islower():
            lower = 1
        elif c.isdigit():
            digit = 1
        elif c in string.punctuation:
            symbol = 1
    score += upper + lower + digit + symbol
    if score <= 2:
        return "Weak"
    elif score <= 4:
        return "Medium"
    else:
        return "Strong"
def generate_password():
    try:
        length = int(entry.get())
        chars = string.ascii_letters + string.digits + string.punctuation
        pwd = ''.join(random.choices(chars, k=length))
        result.set(pwd)
        strength.set(check_strength(pwd))
    except:
        result.set("Invalid input")
        strength.set("")
app = tk.Tk()
app.title("Password Generator & Strength Checker")
tk.Label(app, text="Password Length:").pack(pady=5)
entry = tk.Entry(app)
entry.pack(pady=5)
tk.Button(app, text="Generate Password", command=generate_password).pack(pady=5)
```

```
result = tk.StringVar()
strength = tk.StringVar()
tk.Label(app, text="Generated Password:").pack()
tk.Label(app, textvariable=result).pack()
tk.Label(app, text="Password Strength:").pack()
tk.Label(app, textvariable=strength).pack()
app.mainloop()
```

OUTPUT

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python file_gui.py
```



VOICE USER INTERFACE

PYTHON PROGRAM

```
import speech_recognition as sr
import random
import string
def check_strength(password):
    score = 0
    if len(password) >= 8:
        score += 1
    upper = lower = digit = symbol = 0
```

```
    for c in password:
        if c.isupper():
            upper = 1
        elif c.islower():
            lower = 1
        elif c.isdigit():
            digit = 1
        elif c in string.punctuation:
            symbol = 1
    score += upper + lower + digit + symbol
    if score <= 2:
        return "Weak"
    elif score <= 4:
        return "Medium"
    else:
        return "Strong"
recognizer = sr.Recognizer()
with sr.Microphone() as source:
    print("🎤 Say password length:")
    recognizer.adjust_for_ambient_noise(source)
    audio = recognizer.listen(source)
try:
    recognized_text = recognizer.recognize_google(audio)
    print(f"Google recognized: '{recognized_text}'")
    length = int(recognized_text)
    chars = string.ascii_letters + string.digits + string.punctuation
    password = ''.join(random.choices(chars, k=length))

    print("Generated Password:", password)
    print("Password Strength:", check_strength(password))
```

```
except:
    print("❌ Could not understand the number. Please try again.")
```

OUTPUT

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python file_vui.py
🎤 Say password length:
❌ Could not understand the number. Please try again.
```

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python file_vui.py
🎤 Say password length:
Google recognized: '13'
Generated Password: bUDZDh<Y2k%Qo
Password Strength: Strong
```

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python file_vui.py
🎤 Say password length:
Google recognized: '14'
Generated Password: )]DFT,=Gv<[C5N
Password Strength: Strong
```

USER SATISFACTION COMPARISON

PYTHON PROGRAM

```
print("Please rate your experience with each interface (1 = worst, 5 = best)")
try:
    cli_rating = int(input("CLI Rating: "))
    gui_rating = int(input("GUI Rating: "))
    vui_rating = int(input("VUI Rating: "))
    if any(r < 1 or r > 5 for r in [cli_rating, gui_rating, vui_rating]):
        print("Ratings must be between 1 and 5!")
    else:
        average = (cli_rating + gui_rating + vui_rating) / 3
        print("\n--- Satisfaction Summary ---")
        print(f"CLI Experience: {cli_rating}/5")
        print(f"GUI Experience: {gui_rating}/5")
        print(f"VUI Experience: {vui_rating}/5")
```

```
        print(f"Average Satisfaction Score: {average:.2f}")
        if average >= 4.5:
            print("Excellent! Users are highly satisfied.")
        elif average >= 3.5:
            print("Good! Most users are satisfied.")
        elif average >= 2.5:
            print("Average. There is room for improvement.")
        else:
            print("Low satisfaction. Consider improving the interfaces.")
except ValueError:
    print("Invalid input! Please enter numbers only.")
```

OUTPUT

```
C:\Users\Vedhasree S\Desktop\UI_Comparison_1>python user_satisfaction.py
Please rate your experience with each interface (1 = worst, 5 = best)
CLI Rating: 3
GUI Rating: 5
VUI Rating: 4

--- Satisfaction Summary ---
CLI Experience: 3/5
GUI Experience: 5/5
VUI Experience: 4/5
Average Satisfaction Score: 4.00
Good! Most users are satisfied.
```